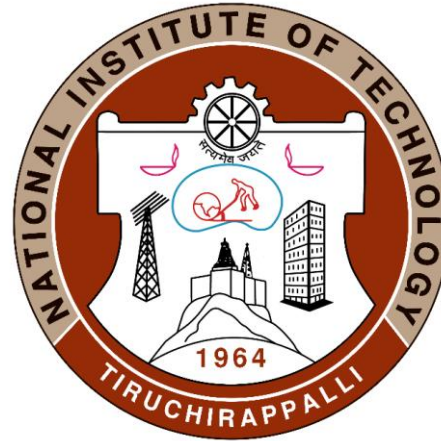


Introduction to Polymers as an Engineering Material



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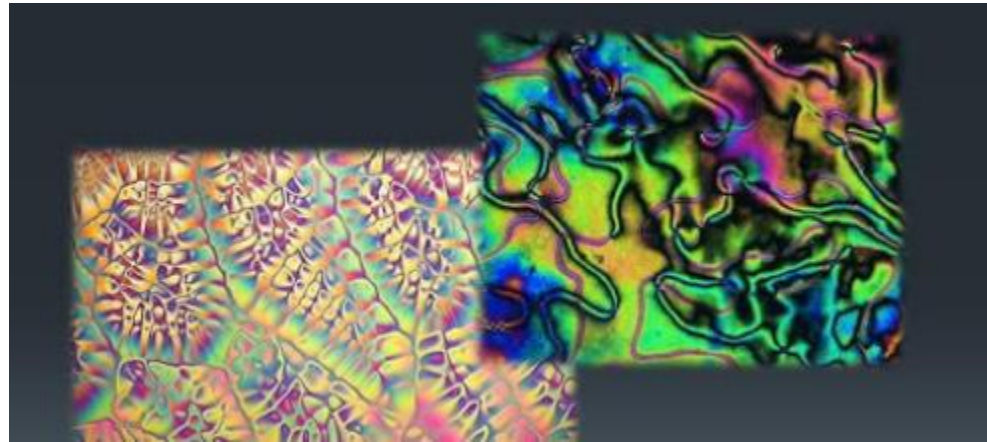
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Lecture - 5

Liquid Crystals



<https://www.youtube.com/watch?v=MuwDwVHVLio>

Contents

1. Introduction
2. What are liquid crystals (LCs)?
3. Classification of liquid crystals
4. Method of characterization of LCs
5. Applications of Liquid Crystals

Introduction

Liquid crystals (LCs) are substances which exhibit the properties of conventional liquid and those of solid crystals

Certain matter which when heated do not directly transform to liquid state but pass through an intermediate state during which the crystal transforms into a cloudy and turbid phase called mesophase

This phase shows the characteristics of flowing like a liquid as well as displaying optical properties of a crystalline solid.

This intermediate state is called mesomorphic state –a region in which properties of liquid and crystalline solid overlap

- The LCs also possess properties not found in either liquids or solids
- May response to external perturbations & some changes color with temperature
- Liquid crystals can flow and be poured into a container.
- The molecules of liquid crystals are essentially rod like so that intermolecular interaction is non-isotropic.
- Mechanically, liquid crystals are highly viscous liquids and display viscous behavior from a honey like state to glassy state

They show large anisodimensionality, i.e., large difference between the intermolecular forces in the length wise direction (++)/-- and in the normal direction (+/-)

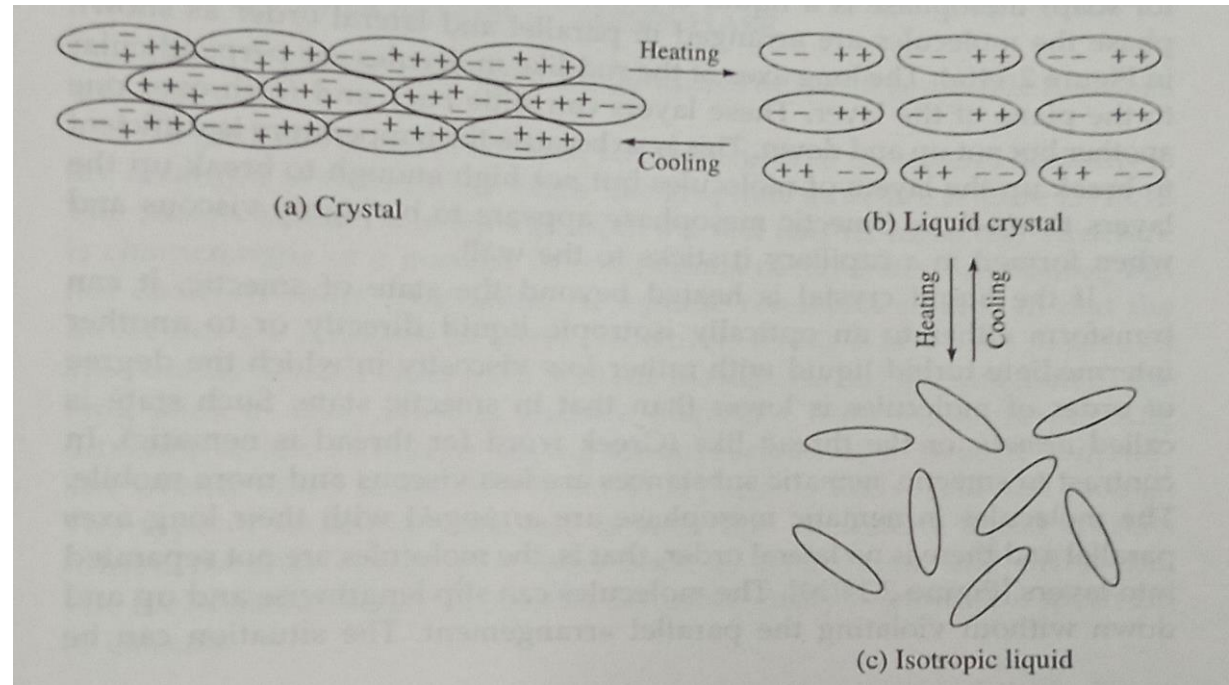


Fig. 1 Molecular organization in the Mesomorphic state

On further heating, the turbid state changes into the isotropic liquid condition. Upon cooling the reaction occurs reversibly

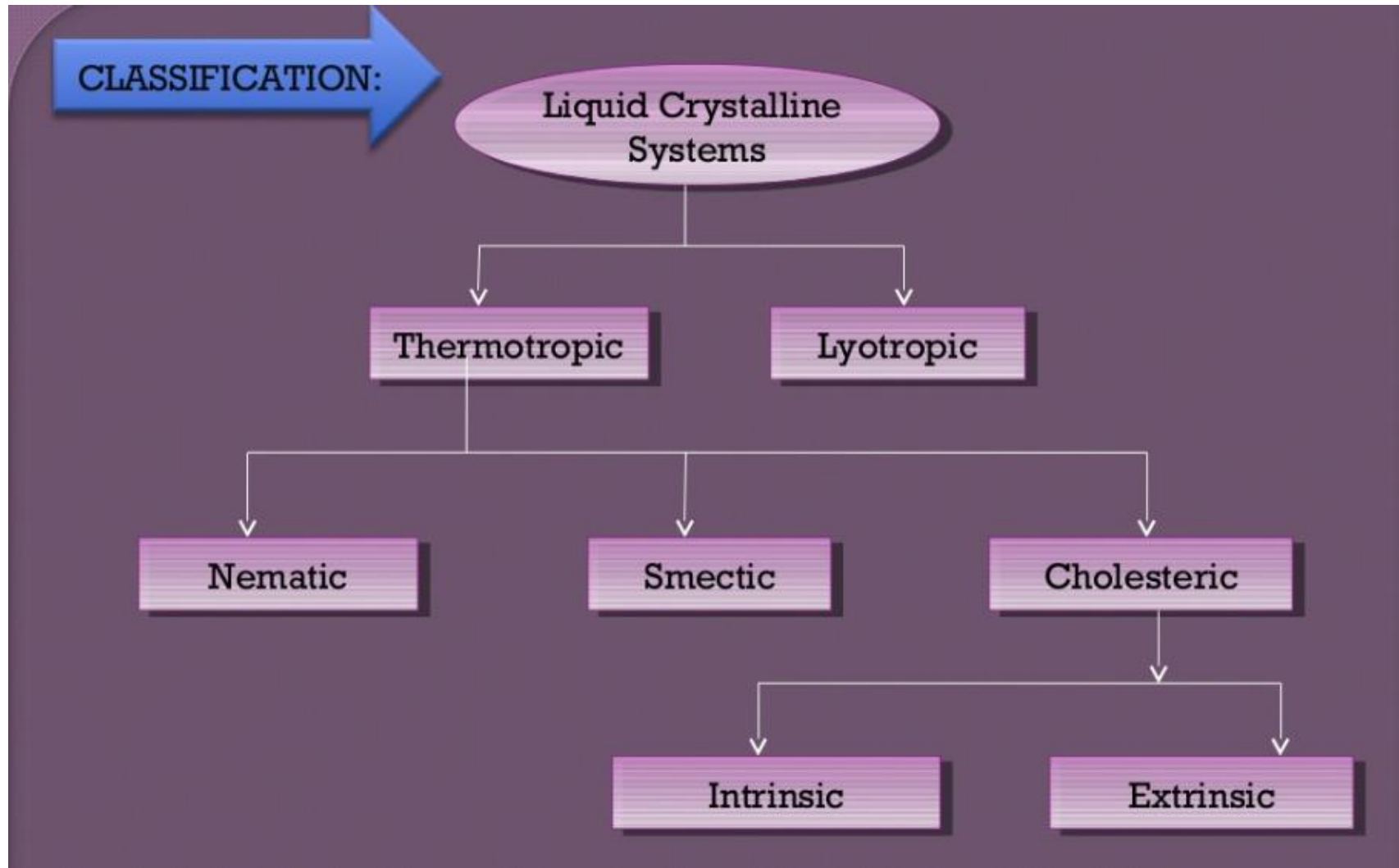


Fig. 2 General Classification of Liquid Crystals

Thermotropic liquid crystals (TLC) are formed during heating of a crystal consisting of rod like molecules in the same way as discussed above.

Lyotropic liquid crystals are formed when certain compounds are treated with a highly polar solvent such as water to form a solution.

In such a case, instead of getting a true liquid phase, a phase characterizing liquid crystal is produced.

An excess of solvent converts the lyotropic mesophase into a true solution. Soaps, various detergents and various biological materials form lyotropic liquid crystals under the influence of solvents.

Three forms of structures have been recognized

1. Smectic
2. Nematic
3. Cholesteric

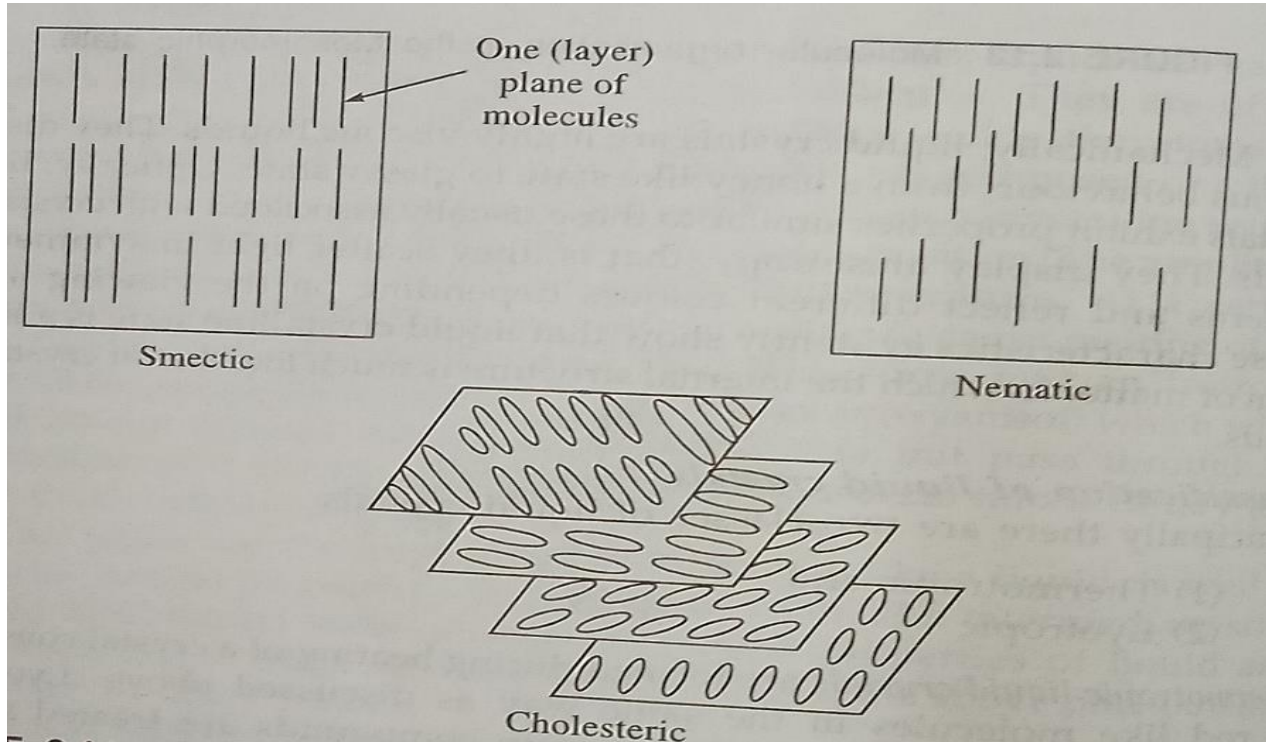


Fig. 3 Molecular arrangement in Smectic, Nematic and Cholesteric Mesophases

Applications of Liquid Crystals

Nematic liquid crystals are transparent in layers thinner than about $25\text{ }\mu\text{m}$. On the application of electric field they turn opaque. This unique optical characteristics makes such substances useful in making display devices such as wrist watch, calculators, etc.

Cholesteric liquid crystals display a number of peculiar optical properties such as polarization and dichroism. They show a variation of color with temperature. Due to this striking characteristics cholesteric liquid crystals find wide applications in making thermal devices

Technological applications

Where are liquid crystals used?

▪ Liquid crystals can be found in the following devices:

- Digital watches
- Pocket TVs
- Gas pumps
- Parking meters
- Telecommunications
- Cell phones and pagers
- High-speed computing
- Digital signs
- Electronic games
- Personal digital assistants
- Electronic books
- Calculators
- Digital cameras and camcorders
- Fishfinders
- Thermometers



Liquid Crystal display devices (LCD)

Both Cholesteric and nematic liquid crystals are employed for making liquid crystals display devices.

LCD can be made by sandwiching a thin film of nematic material between two parallel glass plates

The inner surface of each plate is coated with a transparent and electrically material such as tin oxide or indium tin oxide

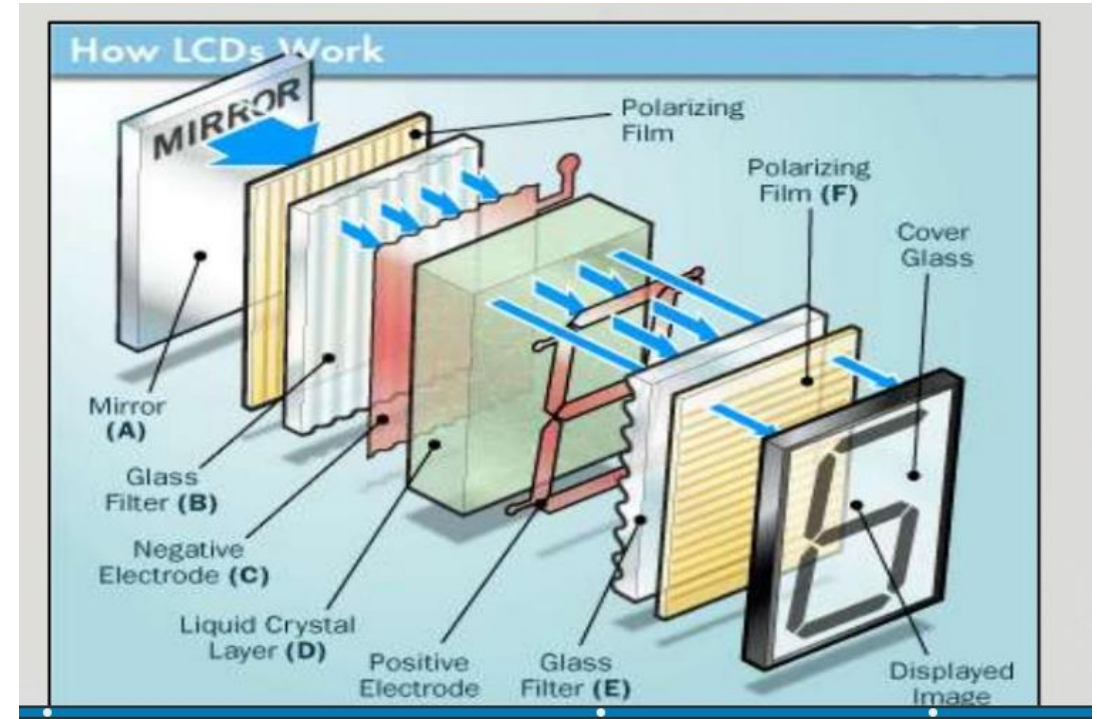


Fig 4. Construction of LCD

- The thickness of the nematic material can vary from 6 to 25 μm .
- Films thinner than $6 \mu\text{m}$ are difficult to make.
- Thicker films cause lot of scattering of light.
- When a small d.c. voltage is applied, the liquid crystal film turns opaque instantaneously due to scattering of light caused by disordered molecules in the nematic liquid crystals. This effect is known as dynamic scattering and is not permanent.
- The clarity of the thin film can be resorted simply by turning off the electric field.

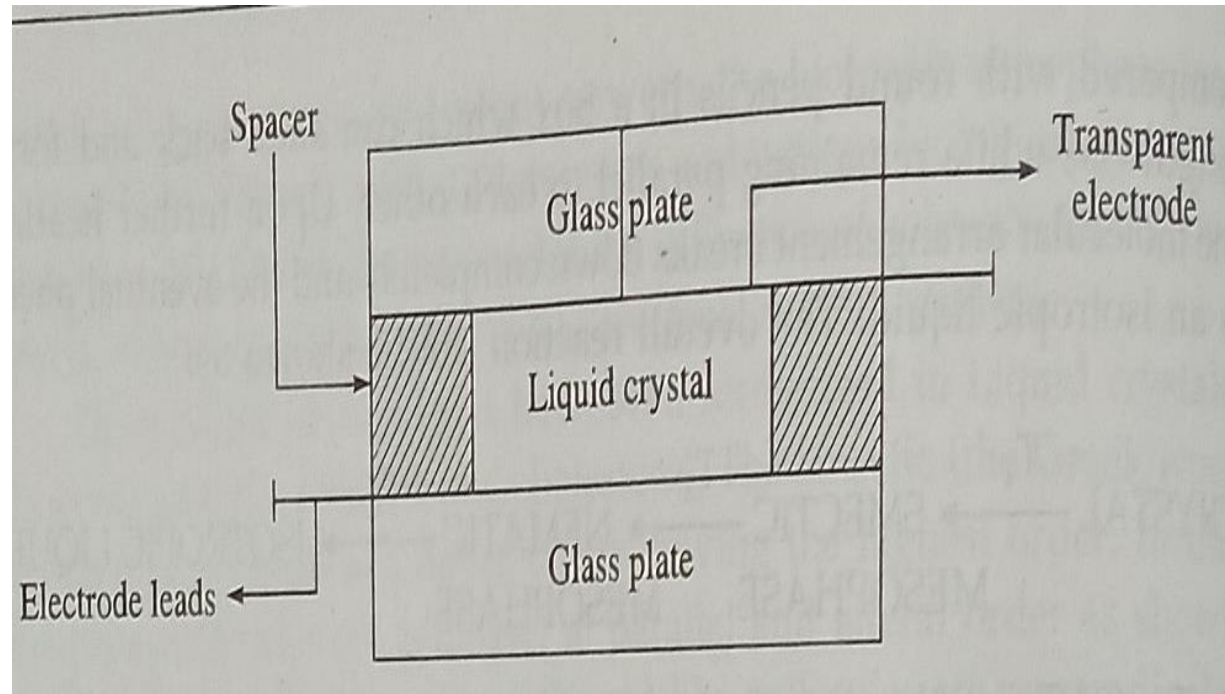


Fig. 5 Cross section of a liquid crystal cell

- If instead of pure nematic crystal, a mixture of nematic and cholesteric liquid crystal is used, the opaqueness can persist on the order of several weeks even if the voltage is turned off.
- On the application of a short pulse of a.c. the material can be made clear. This effect is called storage mode scattering

